

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

An application of uplink RSMA technique for communication of patrol UAVs

HA XUAN SON¹, PHAM THI QUYNH TRANG^{1,2}, LE TRAN MANH¹, and TRINH ANH VU¹

¹Department of Electrical Engineering and Communication, VNU University of Engineering and Technology, Ha Noi, Viet Nam (e-mail: Sonxuanha@yahoo.com.vn, manhlt@vnu.edu.vn, vuta@vnu.edu.vn)

²Faculty of Electrical Engineering and Communication, Ha Noi Industry of University, Viet Nam (e-mail: pham.trang@hau.edu.vn)

Corresponding author: Trinh Anh Vu (e-mail: vuta@vnu.edu.vn).

ABSTRACT Rate-splitting multiple access (RSMA) is gaining significant research attention due to its flexibility in optimizing system capacity and meeting specific application goals. This paper proposes a solution for applying RSMA to transmit data from cameras mounted on unmanned aerial vehicles (UAVs) during patrol missions. Our approach yields deterministic results in the simplest case and can be easily extended to more complex scenarios. Our solution aligns more closely with the intended objectives than the non-orthogonal multiple access (NOMA) method. More specifically, the simulation results show that the proposed technique can achieve high values of achievable rate while still maintaining the power fairness in comparison with conventional techniques. Moreover, our solution has the ability to enhance massive multiple-input multiple-output (MIMO) systems, particularly when multiple user equipments (UEs) share the same spatial angle relative to the base station. In this context, the power domain UE separation technique can effectively complement the spatial angle domain UE separation technique.

INDEX TERMS Rate-splitting multiple access (RSMA), unmanned aerial vehicle (UAV), massive multiple-input multiple-output (mMIMO).

I. INTRODUCTION

THE rapid deployment of fifth-generation (5G) mobile networks has significantly advanced wireless communication capabilities, enabling higher data rates, lower latency, and enhanced connectivity. Since its commercialization in 2019, 5G has continued to attract significant research interest, particularly in optimizing multi-user communication systems for emerging applications such as autonomous surveillance, aerial monitoring, and remote healthcare services [1] [2]. However, as the number of connected devices grows, existing multiple-access techniques face scalability, efficiency, and fairness limitations, necessitating the exploration of more advanced resource allocation strategies.

Non-orthogonal multiple access (NOMA) has been proposed as an enhancement over orthogonal multiple access (OMA), improving spectral efficiency by allowing multiple users to share the same frequency resources through power domain multiplexing. In a NOMA framework, user signals are superimposed at the transmitter and separated at the receiver using successive interference cancellation (SIC) [3] [4]. While NOMA increases capacity, it faces challenges related to inter-user fairness, computational complexity, and degraded performance in high-interference environments.

A more flexible and scalable alternative is rate-splitting multiple access (RSMA), which overcomes the limitations of NOMA by dynamically adjusting the transmission strategy. Unlike NOMA, which operates at peak capacity points within the Gaussian multiple access channel (GMAC), RSMA enables any operating point within the capacity region using a rate-splitting approach. This technique divides original data streams into substreams, which are encoded separately and then sequentially decoded at the receiver [5]–[9]. This adaptive mechanism allows RSMA to manage interference more effectively, optimize spectral efficiency, and ensure a balanced quality of service (QoS) between multiple users.

In a comparison of RSMA, space division multiple access (SDMA), and NOMA schemes, Yijie Mao et al. showed that RSMA provides smoother switching in multi-user, multi-load scenarios with diverse channel directions, strengths, and channel state quality at the transmitter [6]. Additionally, RSMA offers better data rates and quality of service (QoS) than NOMA, while also reducing computational complexity for both the transmission scheduler and receiver (number of SIC layers). Furthermore, the RSMA technique also exhibits greater efficiency and specialization than SDMA, OMA, and NOMA in the sum-rate. This sum-rate is a function of the

difference in channel strengths and the angle between users' channel directions [7]. The RSMA mechanism allows a base station (BS) to adjust the division between common and private messages, managing interference by either decoding it or treating it as noise, to optimize users' data rates. By this division, the RSMA technique achieves up to 15.6% and 21.5% increase in data rate compared to non-orthogonal multiple access (NOMA) and orthogonal frequency division multiple access (OFDMA) [10], [11].

While RSMA has been explored in wireless networks, its application in unmanned aerial vehicle (UAV)-based surveillance and large-scale monitoring remains underdeveloped. This study introduces a novel RSMA-based communication framework tailored for UAV networks, addressing key challenges in multiuser interference management and scalability. The major contributions of this work are:

1) Optimized data stream separation for UAVs: We propose an RSMA-based method to separate data streams from surveillance UAVs operating at equal distances from a control center. This ensures simultaneous and interference-free transmission with minimal computational overhead. More specifically, we first propose a method to separate two data streams from surveillance cameras on two equal-distance UAVs sent simultaneously to the center when they are on patrol duty.

2) Scalability to large-scale UAV networks: Our method extends to M UAVs flying in circular formations, offering a computationally efficient, scalable framework for multi-UAV surveillance. The approach generalizes and simplifies existing models [2], providing a practical formula for real-to-virtual data source transformation. The performance of the proposed framework is simulated and reverified using Monte Carlo simulation, which provides a profound aspect on the sum achievable rate and power fairness of UAVs, demonstrating the advantage over various conventional multiple access techniques.

3) Integration with massive MIMO Systems: RSMA is adapted to complement massive MIMO, particularly when multiple user equipment (UEs) share the same spatial angle relative to a base station (BS). This technique enhances user separation in high-density environments where spatial-domain separation is ineffective.

On the other hand, applications in global healthcare and emergency response, beyond UAV surveillance, have broader implications for real-time healthcare scenarios. Our RSMA-based framework can be applied to such scenarios, such as in telemedicine and remote patient monitoring, where UAVs equipped with medical sensors can transmit critical patient data to hospitals, improving accessibility in remote areas. Also, in medical supply chain optimization, UAVs can enable rapid delivery of essential medical supplies, including vaccines and emergency medications, to underserved regions. Finally, in disaster and emergency response, the proposed framework supports real-time data transmission in disaster zones, ensuring uninterrupted medical communication and coordination.

The remainder of this paper is structured as follows: Sec-

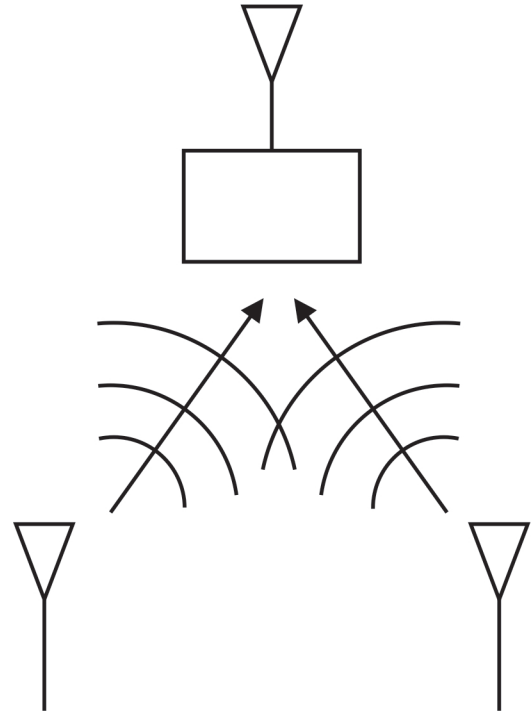


FIGURE 1. Uplink multiple access model of two users.

tion II reviews related work and key challenges in UAV-based communication. Section III presents the proposed RSMA-based model, including optimization strategies and performance evaluation. Section IV discusses scalability, integration with Massive MIMO, and potential applications in global healthcare. Finally, Section V concludes the study and outlines future research directions.

II. RELATED WORDS

The work in [5] presented a scheme to separate two independent uplink sources transmitting to a central receiver.

Fig. 1 shows the uplink multiple-access model of two users. This multiple-access channel is subject to the following constraints [5,12]

$$R_1 < C(P_1, N_0) = \log_2\left(1 + \frac{P_1}{N_0}\right) \quad (1)$$

$$R_2 < C(P_2, N_0) = \log_2\left(1 + \frac{P_2}{N_0}\right) \quad (2)$$

$$R_1 + R_2 = C(P_1 + P_2, N_0) = \log_2\left(1 + \frac{P_1 + P_2}{N_0}\right) \quad (3)$$

where P_1 and P_2 are the received powers at the receiver, and R_1 and R_2 are the rates of the two independent first and second sources, respectively. N_0 is the noise power at the receiver, which we normalize to unity later. Also, C denotes the channel capacity according to the Shannon formula [12], [13], representing the single-channel rate achieved when using random coding. The third equation shows that the

dominant capacity region (the highest total rate) of the system is segments A and B, as illustrated in Fig. 2. Note that the NOMA technique only allows the separation of two UAVs at either point A or B using the SIC method, but not at any point in segment A-B [14]–[16]. In contrast, the RSMA technique allows us to achieve this as in [17]. From (2), we can find a unique value $\delta > 0$ such that

$$R_2 = C(P_2, \sigma^2 + \delta), \quad (4)$$

where $\sigma^2 = N_0$ is noise at the receiver. δ considered as equivalent noise caused by an independent source, it reduces the rate R_2 compared to its single-user capacity $C(P_2, N_0)$.

Based on the value δ , the first source is divided into two virtual sources $P_1 - \delta$ and δ . In total will be three independent sources received at the center, ie. $P_1 - \delta$, P_2 , and δ . For simplicity, we assume the channel gains from transmission to reception are unity, then the power division at the receiver is the same as the power division at the transmitter. Using the SIC method at the receiver with the order of $P_1 - \delta$, P_2 , δ . We obtain the following results

$$R_{11} = C(P_1 - \delta, P_2 + \delta + \sigma^2), \quad (5)$$

$$R_2 = C(P_2, \delta + \sigma^2), \quad (6)$$

$$R_{12} = C(\delta, \sigma^2). \quad (7)$$

It should be mentioned that all three equations can be implemented using single-user coding. Letting $R_1 = R_{11} + R_{12}$, we can easily separate R_1 and R_2 for two UAVs at any points along the segment A-B once the value pair (δ, R_2) is properly selected.

III. UAV COMMUNICATION MODELS PROPOSED FOR RSMA APPLICATION

A. THE MODEL WITH TWO UAVS ON PATROL DUTY

Consider two patrol UAVs flying in circles around a center B. Each UAV is equipped with a camera and transmits video to B in real-time. Then, B is tasked with separating these two data streams to monitor the scene. The following techniques can be implemented:

- If using the OMA technique, the signals of the two data streams from the first and second UAVs are orthogonal to each other, as they are assigned to two different time slots or two frequency bands. Then, B is easy to perform the separation. However, this method would significantly waste the system resources.

- If using the NOMA technique, the data rate separated from two UAVs will be very different, and this scenario would not be meeting the equal-rate and equal-power allocation requirements as two monitoring cameras.

Indeed, let the transmit power of each UAV be P^* . When the signals go through identical channels (with the same distance d), their received powers at B are equal $P = P^*/d^2$. Here, the channel between the UAV and B is assumed to be a line-of-sight (LoS) channel. (The transmission channel with non-line-of-sight (NLoS) probability will be explored in the future).

Applying the data separation technique of NOMA, the rate for UAV1 can be written as

$$R_1 = \log_2(1 + \frac{P}{P+1}). \quad (8)$$

Here the normalized noise power is $N_0 = \sigma^2 = 1$. After using the SIC algorithm, the rate for second UAV can be expressed as

$$R_2 = \log_2(1 + \frac{P}{1}). \quad (9)$$

It is clear that R_1 and R_2 significantly differ with a large range of values $P > 1$. This does not meet the two cameras' service requirement of equal data rates. One solution is to alternate between decoding R_1 and R_2 to achieve equal average rates. However, this solution complicates both control and encoding.

On the other hand, our proposed RSMA-based data separation technique can be applied to this problem as follows. First, split the data stream from the first UAV into two sub-streams corresponding to two power components $P - \delta$ and δ . Then the problem becomes separating three independent sources at B, corresponding to their power and SIC order is $P - \delta$, P , and δ . The SIC algorithm in this order will yield the following results $R_{11} = C(P - \delta, P + \delta + 1)$, $R_2 = C(P, \delta + 1)$, $R_{12} = C(\delta, 1)$, or more explicit this would be expressed as

$$R_{11} = \log_2(1 + (\frac{P - \delta}{P + \delta + 1})), \quad (10)$$

$$R_2 = \log_2(1 + \frac{P}{\delta + 1}), \quad (11)$$

$$R_{12} = \log_2(1 + \frac{\delta}{1}). \quad (12)$$

We can then easily work out $R_1 = R_{11} + R_{12}$. This separation method can achieve the capacity of the multiple-access system with the operating points lying on the segment A-B, including both points A and B of NOMA, as it satisfies the following condition

$$\begin{aligned} R &= R_1 + R_2 = R_{11} + R_{12} + R_2 \\ &= \log_2(1 + \frac{P_1 + P_2}{1}) = \log_2(1 + \frac{2P}{1}). \end{aligned} \quad (13)$$

Unlike [2], where δ is determined by reducing R_1 from its single capacity $C(P_1, 1)$, in our method, δ is determined by the service requirement $R_2 = R_1$, i.e. the data rates from both cameras are equal. Then (13) becomes

$$\begin{aligned} \log_2(1 + \frac{P}{\delta + 1}) &= \log_2(1 + \frac{P - \delta}{P + \delta + 1}) + \log_2(1 + \frac{\delta}{1}) \\ &= \log_2(1 + \frac{2P\delta + P}{P + \delta + 1}), \end{aligned} \quad (14)$$

and then

$$\frac{P}{\delta + 1} = \frac{2P\delta + P}{P + \delta + 1} \longleftrightarrow \frac{1}{\delta + 1} = \frac{2\delta + 1}{P + \delta + 1}. \quad (15)$$

We then have

$$\delta = -1 + \frac{\sqrt{1 + 2P}}{2}. \quad (16)$$

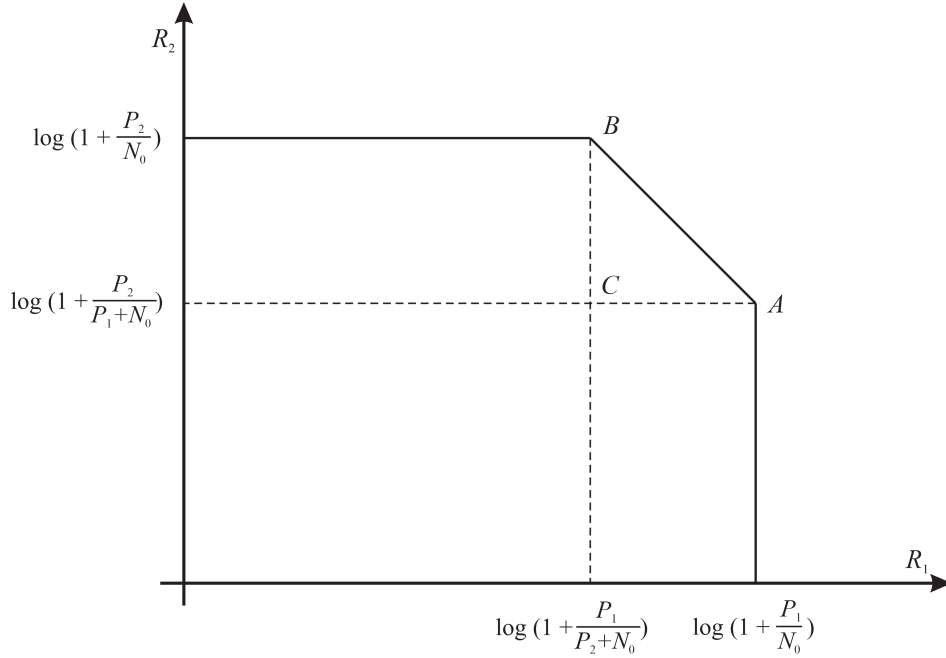


FIGURE 2. The capacity area of two-user multiple access channel.

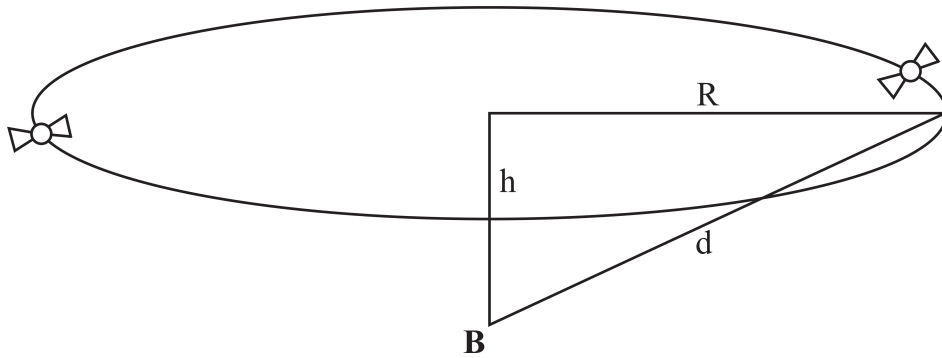


FIGURE 3. Model of two UAVs flying in a circular patrol.

It is easy to see that the solution exists and is satisfied $P > \delta > 0$. That is, it is always possible to divide the first source into two sub-sources to meet service requirements.

Moreover, the solution δ can be found using the bisection algorithm. This method is useful when there are multiple sources and the equation is complex. Consider the system of equations

$$\begin{cases} y_1(\delta) = \log_2(1 + \frac{P-\delta}{P+\delta+1}) + \log_2(1 + \frac{\delta}{1}) \\ y_2 = \frac{1}{2} \log_2(1 + \frac{2P}{1}). \end{cases} \quad (17)$$

It can be seen that y_1 is an a monotonically increasing function of δ because

$$y_{1\delta}' = \log_2 e \left(\frac{1}{\delta+1} - \frac{1}{\delta+1+P} \right) > 0. \quad (18)$$

Besides

$$y_1(0) < \log_2 \left(1 + \frac{P}{P+1} \right) < \log_2 \left(1 + \frac{P}{\delta+1} \right) = y_2$$

,

$$y_1(P) > \log_2 \left(1 + \frac{P}{1} \right) > \log_2 \left(1 + \frac{P}{\delta+1} \right) = y_2.$$

Therefore, when $y_1 = y_2$, the system always has a solution in the interval $(0, P)$, as in Fig. 4. This solution can be found by the bisection Algorithm 1. When normalized by noise, $\sigma^2 = N_0 = 1$, and given that $P = 10$ for example, we would obtain $\delta = 1.79$. The power allocation at the UAVs can be calculated as $\delta^* = 1.79d^2$. That is the power part divided at the transmitter with power $P^* = 10d^2$.

Algorithm 1 Find δ

```

1: begin
2:  $\eta = 0.1$ ;  $\delta_1 = 0$ ;  $\delta_2 = P$ 
3: while  $\delta_2 - \delta_1 > \eta$  do
4:    $\delta = (\delta_2 + \delta_1)/2$ 
5:    $f_1 = |y_1(\delta) - y_2|$ 
6:    $f_2 = |y_1(\delta_2) - y_2|$ 
7:   if  $f_1 f_2 \geq 0$  then
8:      $\delta_2 = \delta$ 
9:   else
10:     $\delta_1 = \delta$ 
11:   end if
12: end while
13: end
14: Output  $\delta$ 

```

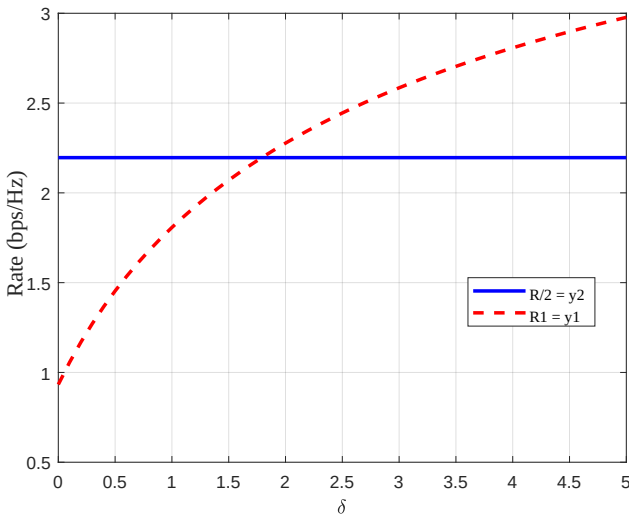


FIGURE 4. Solution by bisection method.

B. THE MODEL WITH THREE UAVS FLYING IN A CIRCLE ($M = 3$)

We now extend the problem to $M = 3$ as three UAVs flying in a circle to monitor the ground with cameras and transmit real-time data to the central station. The service requirement is that all data rates are equal. By splitting the first source into $P - \delta_1$ and δ_1 , the second source into $P - \delta_2$ and δ_2 , and keeping the third source unsplit, we will have a total of $2M - 1 = 5$ virtual sources. Then, with the SIC order $P - \delta_1, P - \delta_2, P, \delta_2, \delta_1$, we obtain following results

$$R_{11} = C(P - \delta_1, P - \delta_2 + P + \delta_2 + \delta_1 + 1) = C(P - \delta_1, 2P + \delta_1 + 1), \quad (19)$$

$$R_{21} = C(P - \delta_2, P + \delta_2 + \delta_1 + 1), \quad (20)$$

$$R_3 = C(P, \delta_2 + \delta_1 + 1), \quad (21)$$

$$R_{22} = C(\delta_2, \delta_1 + 1), \quad (22)$$

$$R_{12} = C(\delta_1, 1). \quad (23)$$

Consequently, all virtual sources above can transmit at capacity-achieving rates with random codes for single-user channels when interference from other sources is considered equivalent noise, independent of synchronization. Then we have

$$R_1 = R_{11} + R_{12}, \quad R_2 = R_{21} + R_{22}, \quad R_3 = R_3. \quad (24)$$

It is clear that $R_1 + R_2 + R_3 = C(3P, 1)$, which is equal to the capacity of the multiple-access system. When there is an additional system requirement of $R_1 = R_2 = R_3$, we have

$$R_1 = C(P - \delta_1, 2P + \delta_1 + 1) + C(\delta_1, 1), \quad (25)$$

$$R_2 = C(P - \delta_2, P + \delta_2 + \delta_1 + 1) + C(\delta_2, \delta_1 + 1), \quad (26)$$

$$R_3 = \frac{1}{3}C(3P, 1). \quad (27)$$

Using the bisection algorithm when combining (25) with (27), we can determine δ_1 . Then we substitute δ_1 into (26), and combine with (27) to solve for δ_2 . With $P = 10$ and $\sigma^2 = N_0 = 1$, the solutions are $\delta_1 = 1.25$ and $\delta_2 = 2.41$ as illustration on the Fig. 5. It can be proven that there

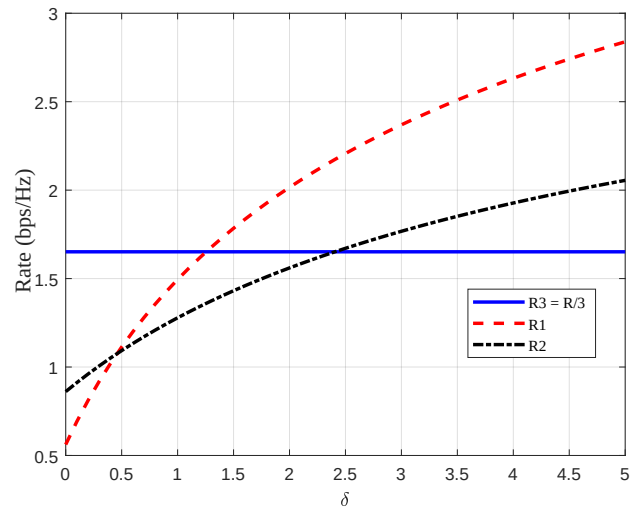


FIGURE 5. Find the solution δ_1 then δ_2 .

always exist unique solutions of the above SIC order since the functions R_1 and R_2 are monotonically increasing to δ . Once δ_1 and δ_2 have been determined, the actual power allocation at the transmitter will be $\delta_1^* = \delta_1 d^2$, $\delta_2^* = \delta_2 d^2$.

C. GENERAL MODEL FOR M UAV FLYING CIRCULAR PATROL

Now we consider the general problem with M UAVs flying in circular patrols. For each data stream from the first $M - 1$ UAV, we divide it into two sub-streams corresponding to the power $P - \delta_i$ and δ_i (where $i = 1, 2, \dots, M - 1$). The M_{th} data stream remains un-split. We have $2M - 1$ independent virtual

sources randomly encoded without requiring strict synchronization. Following a SIC order of $P - \delta_1, P - \delta_2, \dots, P - \delta_{M-1}, P, \delta_{M-1}, \delta_{M-2}, \dots, \delta_2, \delta_1$, we have

$$R_{11} = C(P - \delta_1, P - \delta_2 + \dots + P - \delta_{M-1} + P + \delta_{M-1} + \delta_{M-2} + \dots + \delta_1 + 1) \\ = C(P - \delta_1, (M - 1)P + \delta_1 + 1), \quad (28)$$

$$R_{21} = C(P - \delta_2, P - \delta_3 + \dots + P - \delta_{M-1} + P + \delta_{M-1} + \delta_{M-2} + \dots + \delta_1 + 1) \\ = C(P - \delta_2, (M - 2)P + \delta_2 + \delta_1 + 1), \quad (29)$$

...

$$R_{(M-1)1} = C(P - \delta_{(M-1)}, P + \delta_{M-1} + \delta_{M-2} + \dots + \delta_1 + 1), \quad (30)$$

$$R_M = C(P, \delta_{M-1} + \delta_{M-2} + \dots + \delta_1 + 1), \quad (31)$$

$$R_{(M-1)2} = C(\delta_{M-1}, \delta_{M-2} + \dots + \delta_1 + 1), \quad (32)$$

...

$$R_{22} = C(\delta_2, \delta_1 + 1), \quad (33)$$

$$R_{12} = C(\delta_1, 1). \quad (34)$$

After separating the virtual sources, the rates are combined as $R_i = R_{i1} + R_{i2}$ for $i = 1, 2, \dots, M - 1$. Since the service rates are equal, we have

$$R_1 = R_{11} + R_{12} = \frac{R}{M}, \quad (35)$$

or

$$C(P - \delta_1, (M - 1)P + \delta_1 + 1) + C(\delta_1, 1) \\ = \frac{1}{M}C(MP, 1). \quad (36)$$

From (36) we solve for δ_1 by the bisection algorithm then substitute δ_1 into

$$R_2 = R_{21} + R_{22} = \frac{R}{M}, \quad (37)$$

or

$$C(P - \delta_2, (M - 2)P + \delta_2 + \delta_1 + 1) + C(\delta_2, \delta_1 + 1) \\ = \frac{1}{M}C(MP, 1), \quad (38)$$

we can solve for δ_2 . Continue this process to solve for δ_{M-1} by Algorithm 2.

The result with $P = 10, M = 6, N = 1$ is shown in Fig. 6. The solutions are $\delta = [0.6805, 1.0779, 1.64492, 4.042, 3.3466]$. As in Section A, the existence of a unique solution δ_i in the general case is proven in Appendix A.

Remark: The equal rate when there are M UAVs in multiple access is

$$R_i = \frac{1}{M} \log_2 \left(1 + \frac{MP}{1} \right). \quad (39)$$

Algorithm 2 Find δ_i

```

1: Input:  $M, P, a = a + \delta_i$ 
2:  $y_2 = \frac{1}{M} \log_2(1 + MP)$ 
3:  $\eta = 0.001; \delta_1 = 0; \delta_2 = P$ 
4: while  $\delta_2 - \delta_1 > \eta$  do
5:    $\delta = (\delta_2 + \delta_1)/2$ 
6:    $f_1 = \log_2 \left( 1 + \frac{P - \delta}{(MP + \delta + a + 1)} \right)$ 
        $+ \log_2 \left( 1 + \frac{\delta}{a + 1} \right) - y_2$ 
7:    $f_2 = \log_2 \left( 1 + \frac{P - \delta_2}{(MP + \delta_2 + a + 1)} \right)$ 
        $+ \log_2 \left( 1 + \frac{\delta_2}{(a + 1)} \right) - y_2$ 
8:   if  $f_1 f_2 \geq 0$  then
9:      $\delta_2 = \delta$ 
10:  else
11:     $\delta_1 = \delta$ 
12:  end if
13: end while
14: return  $\delta_1$ 
15: Output  $\delta_1$ 

```

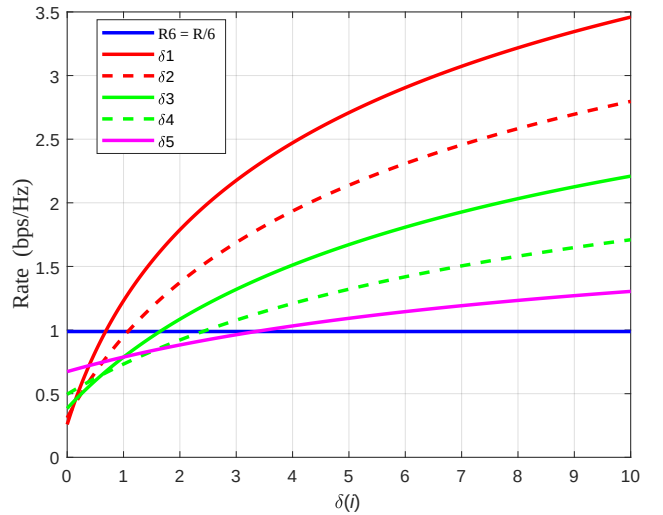


FIGURE 6. Solution of δ_i ($i = 1, \dots, 5$) for $M = 6$.

This rate decreases monotonically with M when P is fixed since the first derivative of $R < 0$ as in Fig. 6. When M is large enough, the data rate from the UAV is smaller than the service rate, such as real-time video transmission. Then, additional bandwidth B (Hz) can be provided. However, since the bandwidth B allocated to the service is usually predetermined, the maximum number of UAVs, M , is determined by Algorithm 3

D. GENERAL PATROL UAV MODEL AND APPLICATION TO MASSIVE MIMO SYSTEMS

In this section, we consider M patrol UAVs at different distances to the central hub B. To turn this problem into that introduced in Section 3, we simply add a power control loop

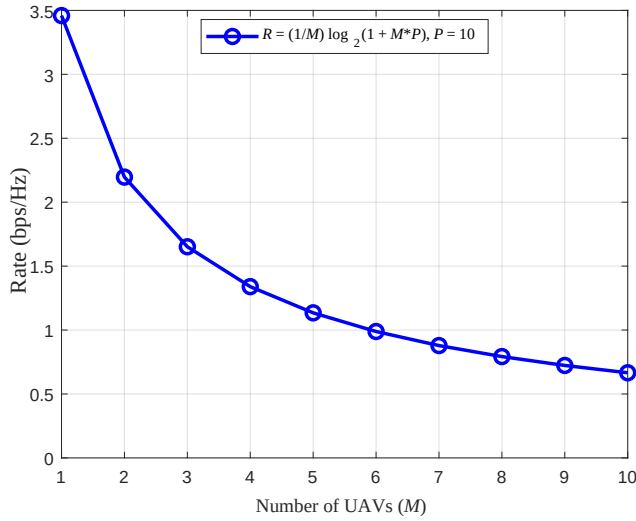


FIGURE 7. The data rate of each UAV decreases as the number of UAVs participating in multiple access is M increases.

Algorithm 3 Find the maximum number of UAVs

```

1: Input:  $B$  (Hz),  $P$  (W) =  $P^*/d^2$ ,  $Q$  (bps, the required data
   transmission rate from the camera)
2: Maximum  $M$ 
3: Subject to:  $Q \leq B \frac{1}{M} \log_2(1 + MP)$ 
4: for  $t = 1 : M - 1$  do
5:    $\eta = 0.1$ ;  $\delta_1 = 0$ ;  $\delta_2 = P$ 
6:    $y_M = \frac{B}{M} \log_2(1 + MP)$ 
7:   while  $\delta_2 - \delta_1 > \eta$  do
8:      $\delta = (\delta_2 + \delta_1)/2$ 
9:      $f_1 = B \log_2(1 + \delta) - y_M$ 
10:     $f_2 = B \log_2(1 + \delta_2) - y_M$ 
11:    if  $f_1 f_2 \geq 0$  then
12:       $\delta_2 = \delta$ 
13:    else
14:       $\delta_1 = \delta$ 
15:    end if
16:  end while
17:   $\delta(t) = \delta$ 
18: end for

```

as follows

- At each coherent time interval, the central hub B will transmit a common pilot signal to all UAVs. Each UAV will then measure the channel gain to it and adjust its transmission power to be inversely proportional to the channel gain to the central hub as $P^* = \frac{P_0}{\gamma} < P_{max}$. Here, P_0 is the pre-selected appropriate value.
- The central hub B not only receives the same power P from each UAV but can also estimate the number of UAVs, and M from the sum of these signals. B then transmits the value of M back to the UAVs, and based on this information, the UAVs will divide their data streams into sub-streams as described in Section 3.

At the end of this paper, we propose to apply the above-mentioned RSMA-based technique to the massive MIMO system to add the ability to separate UEs that have the same spatial angle to the BS. Use the results in [18]–[21], after processing by the maximum-ratio combining (MRC) technique. We have

$$SE_k^{LoS} = \log_2 \left(1 + \frac{M}{\beta g(\phi_k^0, \phi_{k+1}^0) + \frac{1}{SRN_k}} \right). \quad (40)$$

Here, the left side denotes the spectral efficiency of k -th UE interfered by $k+1$ -th UE at the same frequency in the LOS channel. On the right side, M is the number of antennas at the BS, ϕ_k^0, ϕ_{k+1}^0 are the spatial angles of k -th UE and $k+1$ -th UE to the BS, SNR is the signal-to-noise ratio, and β is the gain ratio of the two channels. If we have $\sin \phi_k^0 \neq \sin \phi_{k+1}^0$ and $\beta g(\phi_k^0, \phi_{k+1}^0)$ decrease as $\frac{1}{M}$. That means with M is large, $k+1$ -th UE is almost orthogonal to k -th UE if their spatial angles to BS are different.

However when $\sin \phi_k^0 = \sin \phi_{k+1}^0$, $g(\phi_k^0, \phi_{k+1}^0) = M$ $k+1$ -th UE causes strong interference to k -th UE, make difficult to separate k -th UE and $k+1$ -th UE. In this case, the RSMA-based separation technique presented above can be properly applied. In the first phase, BS estimates the channels from the orthogonal pilots transmitted from the UEs. After processing according to MRC, we have

- If the SNR value is greater than the threshold value η . BS separates the desired UE signal normally using massive MIMO technique.
- If the SNR value is smaller than the threshold η . This means there are many UEs with similar directional vectors towards the BS. In this case, the BS will perform beamforming using a pilot in this direction so that all UEs in this direction can estimate the channel gain to them.

In the next step these UEs will transmit a signal with a power inverse to their channel gain. The BS determines the number UEs in this direction as M and broadcasts it to the UEs. When each UE knows the number M in the same direction, it will automatically divide its data stream into corresponding virtual sources as described in Section 3. Finally, the BS can separate UEs that have the same spatial direction.

Remark: According to the requirements of the massive MIMO system, we should have $M \gg K$ where M represents the number of base station antennas and K is the number of UEs in a cell to ensure good signal separation with distinct spatial angles from UEs to the BS [22]. With the addition of RSMA to the signal separation in the power domain for a group of UEs with the same spatial direction towards the BS, K can be increased, the spectral efficiency of massive MIMO will further improve.

IV. SIMULATION RESULTS AND DISCUSSION

In this Section, performance comparison of OMA, NOMA, and the proposed scheme for different numbers of UAVs is demonstrated through Monte Carlo simulations. In the figures, we consider the total achievable rate and fairness metrics

for each scheme, where equal achievable rates among UAVs are ensured to maintain fairness in user allocation. Also, we define the power fairness index (PFI) as a crucial metric for evaluating the fairness of power allocation among UAVs. More specifically, PFI measures the degree to which power is distributed equally across UAVs. Therefore, this parameter can help to ensure a balanced allocation. The PFI value ranges from zero to one, where a value of one represents an ideal case of equal power allocation among all UAVs, and values approaching zero indicate a highly unfair power distribution. PFI can be defined as

$$PFI = \frac{\left(\sum_{i=1}^N P_i\right)^2}{N \sum_{i=1}^N P_i^2}, \quad (41)$$

where P_i is the allocated power for i -th UAV, and N denotes the total number of UAV in the system. In here, the numerator corresponds to the squared sum of all allocated power values, while the denominator is the sum of the squared power allocations. Then the result is normalized by the number of UAVs in the system.

Moreover, maintaining a high value of PFI is crucial in practical multi-user communication networks, especially in UAV-enabled communications or IoT networks. Therefore, fair power allocation improves system performance would extends device lifetime, helping to ensure stable connectivity. If power is distributed uniformly among all devices, the numerator and denominator result in a maximum PFI value of one, indicating perfect fairness. In contrast, if power is unevenly distributed between UAVs and some UAVs receive significantly more power than others, decrease the PFI value to zero. For example, in a severely imbalanced situation where one UAV receives most of the power while other UAVs receive only an insignificant power allocation, the PFI will be near zero, indicating an unfair distribution of power.

In Fig. 8, the performance of OMA, NOMA, and the pro-

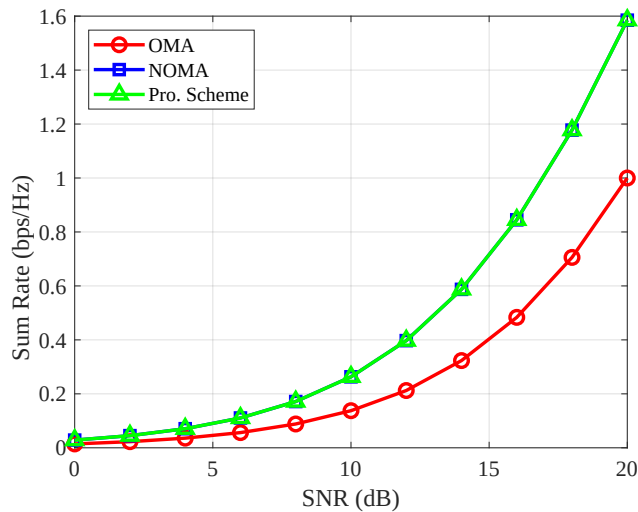


FIGURE 8. Sum rate comparison of two UAVs scenario.

posed scheme is compared under equal rate conditions. It can be seen from Fig. 8 that the total rate of the proposed scheme and NOMA are the same and always higher than that of the OMA at all SNR values, indicating the superior performance of the proposed scheme under high SNR conditions. On the other hand, the total rates of all three techniques are almost the same at low SNR. As the SNR increases, the gap between the output of OMA, NOMA, and the proposed scheme becomes larger. When the SNR nears the value of 20 dB, the total rate of the proposed scheme is significantly higher than that of OMA, demonstrating the significant benefit of this method in improving the system performance.

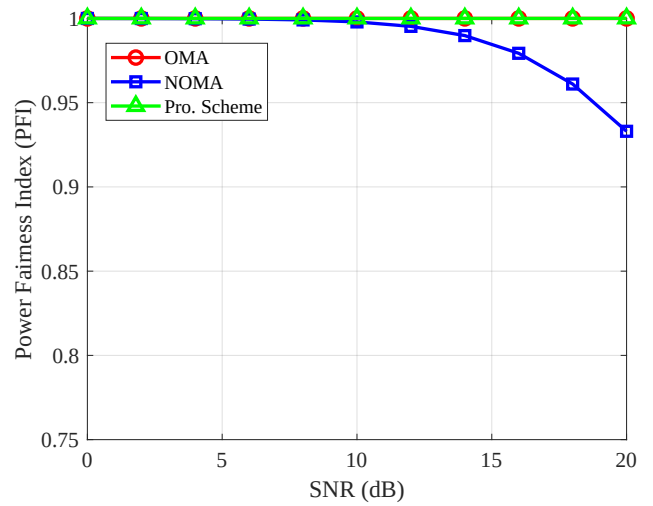


FIGURE 9. PFI comparison of two UAVs scenario.

On the other hand, the PFI values of all schemes are compared in Fig. 9. In this scenario, the power allocation of the NOMA scheme is unequally set to ensure uniform rates between two UAVs. It can be seen from the figure that the power fairness index of the proposed scheme and OMA always remains close to one, whereas the PFI of NOMA decreases significantly as the SNR increases. At low SNR, all three methods have nearly equal PFI, but as the SNR increases, the PFI of NOMA begins to decrease significantly, indicating the deterioration of power fairness under high SNR channel conditions. When the SNR reaches 20 dB, the PFI of NOMA drops below 0.94, while the proposed method still maintains the maximum, demonstrating the superior power fairness maintenance ability of this scheme.

In Fig. 10, the rate performance is presented for the three-UAVs scenario, demonstrating the significant performance gains of the proposed scheme. The power fairness index is also examined, revealing the fairness limitations of NOMA and highlighting the advantage of the proposed scheme in maintaining equal power distribution among users. Similar to the previous case, achievable rates of all UAVs are equal to ensure fairness between users. It can be seen from the figure that the total rate of NOMA and the proposed scheme is significantly higher than that of OMA over the entire

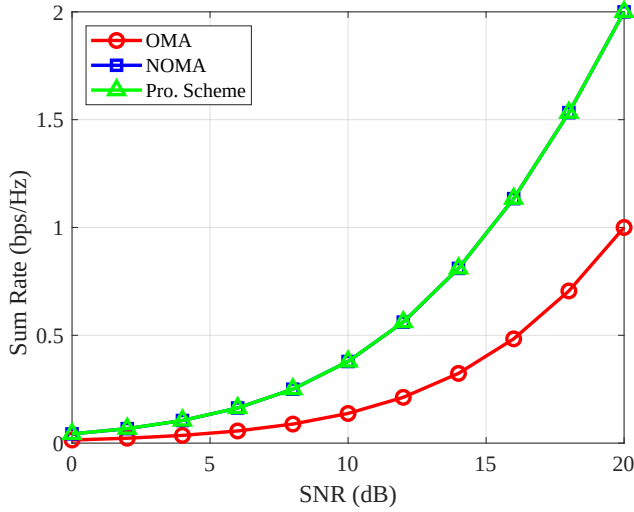


FIGURE 10. Sum rate comparison of three UAVs scenario.

SNR range, demonstrating the superior performance of this method. At low SNR, the total rates of all three techniques are almost similar. As the SNR increases, the proposed RSMA-based technique begins to show a clear advantage over the other methods. When the SNR reaches 20 dB, the total rate of the proposed method is significantly larger than that of OMA, confirming the effectiveness of the proposed scheme in improving the system capacity.

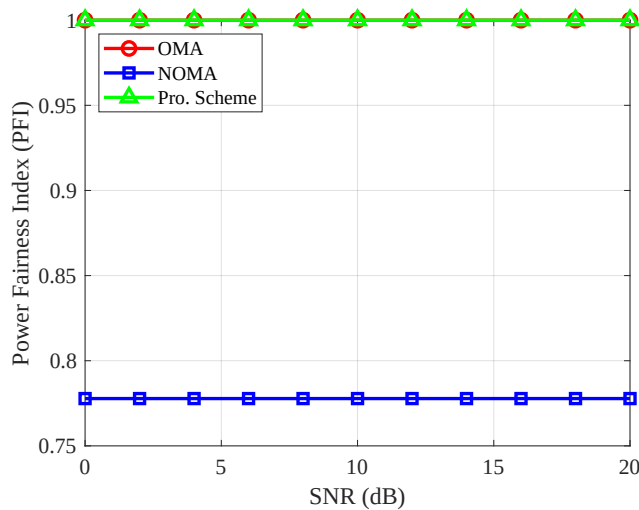


FIGURE 11. PFI comparison of three UAVs scenario.

Similar to the case of two UAVs, it can be seen from Fig. 11 that the power fairness of the proposed scheme and OMA always remains at the maximum level of one over the entire SNR range, while NOMA has a significantly lower PFI. For the NOMA method, the PFI fluctuates around 0.78 and does not change significantly with SNR, indicating unfairness in power allocation among users to ensure equal rate conditions. In contrast, the proposed scheme achieves optimal fairness,

equal to OMA and significantly outperforms NOMA, confirming its effectiveness in fair power allocation among users.

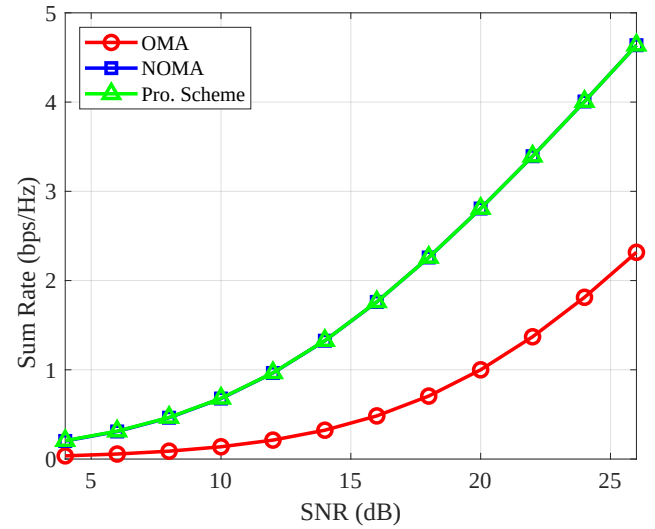


FIGURE 12. Sum rate comparison of six UAVs scenario.

Fig. 12 presents the sum rate performance of different schemes as a function of SNR for the case of six users. In accordance with the previous results, we confirm that the proposed scheme outperforms OMA performance. We observe that the proposed scheme consistently achieves the highest rate across all SNR values, demonstrating a significant improvement over OMA and a slight edge over NOMA in low-SNR regimes. The rate gain of the proposed scheme over OMA is particularly notable, confirming the efficiency of resource allocation in the proposed method. As shown in Fig. 13, the advantage of the proposed scheme becomes more evident at higher SNR, but the proposed scheme still offers superior performance due to its improved power allocation strategy.

In contrast, the PFI performance of six UAVs in Fig. 13 shows the inferiority of NOMA in comparison with OMA and the proposed scheme. As expected, the proposed scheme maintains an optimal fairness level similar to OMA, while NOMA exhibits a significantly lower fairness index. More specifically, both the proposed scheme and OMA achieve a PFI close to one across all SNR values, indicating that they ensure a fair power allocation among users. In contrast, NOMA consistently maintains a PFI around 0.5, reflecting an inherent fairness issue in its power allocation strategy. Therefore, even though the proposed scheme has similar rate performance to NOMA, the fairness advantage of the proposed scheme over NOMA is evident, confirming its capability to balance power distribution effectively. The power balance strategy is critically important in the UAV patrol scenarios, where the duty time of each UAV should be equally maintained to ensure continuous surveillance.

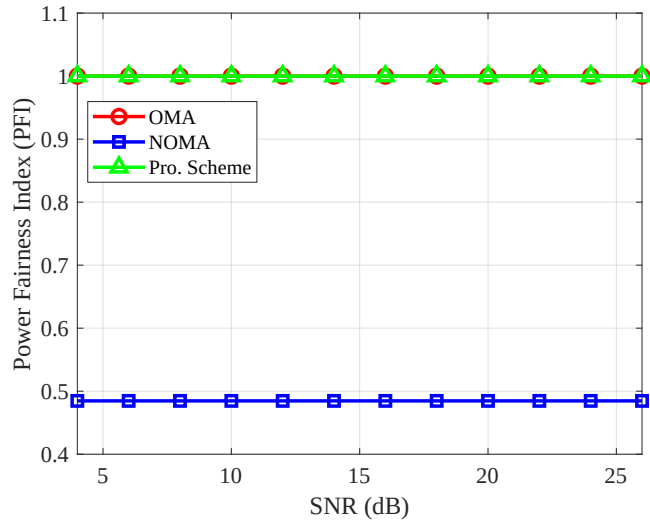


FIGURE 13. PFI comparison of six UAVs scenario.

V. CONCLUSION

This paper presents the application of RSMA for data transmission from cameras mounted on patrol UAVs to a central hub. Unlike [5], where power is divided from a single source into virtual sources based on how much the rate is reduced compared to the single-user capacity, in this paper, the power division is standardized based on equal data rates from the UAVs. Using a chosen SIC order, this paper describes a simple algorithm for calculating the power allocation values for RSMA in the general case of M UAVs without proof by induction, as in [5]. Furthermore, this power allocation method for RSMA can be extended to UAVs operating at arbitrary distances to the central hub by adding a simple power control loop. This technique complements massive MIMO systems by separating UEs with the same spatial angle to the BS using the power domain. Simulation results show that our proposed technique is a well-balanced solution that not only achieves high rate but also maintains a fair power allocation, making it a promising approach for UAV networks. Also, the increased computational complexity of RSMA decoding can challenge resource-constrained UAV systems. The SIC process in RSMA requires additional computational complexity compared to OMA, which could impact real-time applications. Furthermore, RSMA performance largely depends on perfect channel state information (CSI) as channel estimation errors can degrade performance in practical UAV deployments. Therefore, it is essential to develop robust CSI acquisition techniques to facilitate our proposed technique. Future research will further explore the channel model with LoS probability between UAVs and the BS. Another important direction is considering adaptive schemes that can dynamically change between multiple access techniques based on real-time channel conditions. Also, our proposed technique can be integrated with machine learning-based frameworks to further improve its adaptability in UAV communication

environments.

APPENDIX A

To prove that $\delta_i, i = 1, 2, \dots, M-1$ exists uniquely on the interval $(0, P)$, we first prove that $R_i(\delta_i)$ is an increasing monotone function to δ_i . In here

$$R_i(\delta_i) = C(P - \delta_i, (M-i)P + \delta_i + \delta_{i-1} + \dots + \delta_1 + 1) + C(\delta_i, \delta_{i-1} + \dots + \delta_1 + 1), \quad (42)$$

or rewrite

$$R_i(\delta_i) = \log_2\left(1 + \frac{P - \delta_i}{(M-i)P + \delta_i + a}\right) + \log_2\left(1 + \frac{\delta_i}{a}\right). \quad (43)$$

Here $a = \delta_{i-1} + \dots + \delta_1 + 1$ is the sum of previously determined δ_j values ($j < i$). Then we have

$$R_i(\delta_i) = \log_2 \exp\left(\ln\left(\frac{(M-i+1)P + a}{((M-i)P + \delta_i + a)}\right) + \ln\left(\frac{\delta_i + a}{a}\right)\right), \quad (44)$$

$$R'_i(\delta_i) = \log_2 \exp\left(\frac{-1}{(M-i+1)P + \delta_i + a} + \frac{1}{(\delta_i + a)}\right). \quad (45)$$

Clearly $R'_i(\delta_i) > 0$ when $1 < i < M-1$. Besides, we have

$$\begin{aligned} R_i(0) &= \log_2\left(1 + \frac{P}{((M-i)P + a)}\right) < R_M \\ &= \log_2\left(1 + \frac{P}{a}\right) = \frac{1}{M} C(MP, 1), \end{aligned} \quad (46)$$

and

$$R_i(P) = \log_2\left(1 + \frac{P}{a}\right) = R_M = \frac{1}{M} C(MP, 1). \quad (47)$$

This means that δ_i exists uniquely in the interval $(0, P)$.

REFERENCES

- [1] IMT-2020 (5G) Promotion Group, "5G Vision and Requirements," 2014.
- [2] Thompson, J., et al., "5G wireless communication systems: prospects and challenges," *IEEE Communications Magazine*, vol. 52, no. 2, pp. 62-64, 2014. (Guest Editorial).
- [3] Benjebbour, A., Saito, Y., Kishiyama, Y., Li, A., Harada, A., Nakamura, T., "Concept and practical considerations of non-orthogonal multiple access (NOMA) for future radio access," in *Proc. Int. Symp. Intelligent Signal Processing and Communications Systems (ISPACS)*, Naha, Japan, 2013, pp. 770-774.
- [4] Higuchi, K., Benjebbour, A., "Non-orthogonal multiple access (NOMA) with successive interference cancellation for future radio access," *IEEE Transactions on Communications*, vol. 98, no. 3, pp. 403-414, 2015.
- [5] B. Rimoldi and R. Urbanke, "A rate-splitting approach to the Gaussian multiple-access channel," *IEEE Transactions on Information Theory*, vol. 42, no. 2, pp. 364-375, Mar. 1996.
- [6] Y. Mao, B. Clerckx, and V. O. K. Li, "Rate-splitting multiple access for downlink communication systems: Bridging, generalizing, and outperforming SDMA and NOMA," *EURASIP Journal on Wireless Communications and Networking*, vol. 2018, no. 133, 2018. DOI: 10.1186/s13638-018-1104-7.
- [7] B. Clerckx, Y. Mao, R. Schober, and H. V. Poor, "Rate-splitting unifying SDMA, OMA, NOMA, and multicasting in MISO broadcast channel: A simple two-user rate analysis," *IEEE Wireless Communications Letters*, vol. 9, no. 3, pp. 349-353, Mar. 2020.

- [8] Z. Yang, M. Chen, W. Saad, and M. Shikh-Bahaei, "Optimization of rate allocation and power control for rate-splitting multiple access (RSMA)," *IEEE Transactions on Communications*, vol. 69, no. 9, pp. 6013-6027, Sep. 2021.
- [9] A. Kumar, F. Y. Li, and J. Martinez-Bauset, "Revealing the benefits of rate-splitting multiple access for uplink IoT traffic," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2022.
- [10] Y. Mao, O. Dizdar, B. Clerckx, R. Schober, and P. Popovski, "Rate-splitting multiple access: Fundamentals, survey, and future research trends," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 1854-1897, Fourth Quarter 2022.
- [11] Saito, Y., Kishiyama, Y., Benjebbour, A., Nakamura, T., Li, A., and Higuchi, K., "Non-orthogonal multiple access (NOMA) for cellular future radio access," in *Proc. IEEE 77th Vehicular Technology Conference (VTC Spring)*, Dresden, Germany, 2013.
- [12] T. Cover and J. Thomas, *Elements of Information Theory*, 2nd ed., Wiley-Interscience, New York, 2006.
- [13] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*, Cambridge University Press, Cambridge, UK, 2005.
- [14] H. Joudeh, B. Clerckx, and C. Xu, "Sum-rate maximization for linearly precoded downlink multiuser MISO systems with partial CSIT: A rate-splitting approach," *IEEE Transactions on Communications*, vol. 64, no. 11, pp. 4847-4861, Nov. 2016.
- [15] H. Joudeh and B. Clerckx, "Robust transmission in downlink multiuser MISO systems: A rate-splitting approach," *IEEE Transactions on Signal Processing*, vol. 64, no. 23, pp. 6227-6242, Dec. 2016.
- [16] Y. Mao, B. Clerckx, and H. V. Poor, "Rate-splitting for multi-antenna non-orthogonal unicast and multicast transmission," in *Proc. IEEE Signal Processing Advances in Wireless Communications (SPAWC)*, 2018.
- [17] O. Dizdar, Y. Mao, Y. Xu, P. Zhu, and B. Clerckx, "Rate-splitting multiple access for enhanced URLLC and eMBB in 6G," in *Proc. IEEE International Symposium on Wireless Communication Systems (ISWCS)*, Oct. 2021, pp. 1-6.
- [18] Y. Mao, E. Piovano, and B. Clerckx, "Rate-splitting multiple access for overloaded cellular Internet of Things," *IEEE Transactions on Communications*, vol. 69, no. 7, pp. 4504-4519, Jul. 2021.
- [19] Z. Yang, M. Chen, W. Saad, and M. Shikh-Bahaei, "Optimization of rate allocation and power control for rate-splitting multiple access (RSMA)," *IEEE Transactions on Communications*, vol. 69, no. 9, pp. 6013-6027, Sep. 2021.
- [20] T. L. Marzetta, "Noncooperative cellular wireless with unlimited numbers of base station antennas," *IEEE Transactions on Wireless Communications*, vol. 9, no. 11, pp. 3590-3600, Nov. 2010.
- [21] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, "Massive MIMO for next generation wireless systems," *IEEE Communications Magazine*, vol. 52, no. 2, pp. 186-195, Feb. 2014.
- [22] H. Q. Ngo, E. G. Larsson, and T. L. Marzetta, "Energy and spectral efficiency of very large multiuser MIMO systems," *IEEE Transactions on Communications*, vol. 61, no. 4, pp. 1436-1449, Apr. 2013.



HA XUAN SON received the B.S. and M.S. degrees from VNU University of Engineering and Technology, in 2020 and 2022, respectively. He is currently pursuing a Ph.D. degree in telecommunication engineering at VNU. He is interested in wireless communication, wireless sensor network Optimization algorithms, digital signal processing, neural networks, nature-inspired algorithm applications, and FPGA technology.



PHAM THI QUYNH TRANG received the B.S. and M.S. degrees from VNU University of Engineering and Technology, in 2003 and 2006, respectively. She is currently pursuing a Ph.D. degree in telecommunication engineering at VNU. She is interested in wireless communication, wireless sensor network Optimization algorithms, digital signal processing, neural networks, nature-inspired algorithm applications, and FPGA technology.



LE TRAN MANH was an Assoc. Prof. at Wireless Communication Department of VNU University of Engineering and Technology. His main research interests are High-speed Transmission in Wireless Communications, Massive MIMO systems for 5G, Millimeter wave Communications, FPGA Design of Communication System.



TRINH ANH VU was an Assoc. Prof. at Wireless Communication Department of VNU University of Engineering and Technology. His main research interests are High-speed Transmission in Wireless Communications, Massive MIMO systems for 5G, Millimeter wave Communications, FPGA Design of Communication System.

...